Vanilla Options in Practice

Lecture 9 Notes — Payoffs, Black–Scholes Benchmarks, and Static-Arbitrage Diagnostics

Goals and logical structure

Lecture 8 developed model-free no-arbitrage principles. This lecture applies them to the option quotes observed in a market:

$$\begin{aligned} \text{option payoffs} &\longrightarrow \text{replication and put–call parity} \\ &\longrightarrow \text{Black–Scholes benchmark prices} \\ &\longrightarrow \text{model-free bounds and strike-shape checks} \\ &\longrightarrow \text{explicit arbitrage trades.} \end{aligned}$$

The recurring method is always the same: write the terminal payoff of a portfolio, compare it state by state, then translate the comparison into a price inequality or a trading strategy.

1 Payoffs and market language

1.1 European calls and puts

Definition 1.1 (European vanilla options)**.** Let S_T be the price of an underlying asset at maturity T, and let K be a strike price. A European call gives its holder the right to buy one share for K at T; a European put gives its holder the right to sell one share for K at T. Their terminal payoffs are

$$\boxed{(S_T - K)^+} \qquad \text{and} \qquad \boxed{(K - S_T)^+},$$

respectively. European means that exercise is possible only at T.

The option holder chooses whether to exercise. For a call, paying K to receive a share is useful only when $S_T > K$; for a put, selling a share for K is useful only when $S_T < K$. A short position has the negative of the corresponding payoff.

Example — reading a payoff table

For $K = 100$,

S_T	80	100	130
$(S_T - 100)^+$	0	0	30
$(100 - S_T)^+$	20	0	0

The call is upside exposure. The put is downside insurance.

1.2 Moneyness and time value

Moneyness compares the current spot price S_t with the strike K:

	in the money	at the money	out of the money
call	$S_t > K$	$S_t \approx K$	$S_t < K$
put	$S_t < K$	$S_t \approx K$	$S_t > K$.

It is a statement about immediate exercise value, not a forecast of whether the option will finish in the money at T. For a call, the intrinsic value is $(S_t - K)^+$; for a put it is $(K - S_t)^+$. The remaining part of the market price is its *time value*. At-the-money options often have the greatest time value because their intrinsic value is near zero while there is still substantial scope for a favourable move.

Exercise — which contract is mostly time value?

Three calls have the same maturity and current spot $S_t = 100$, but strikes 80, 100, and 130. Which call is most naturally interpreted as mostly time value?

Short solution

The call struck at 100 is at the money, so its intrinsic value is zero. Its price is principally the value of the uncertainty that remains before maturity. The $K = 80$ call already has intrinsic value; the $K = 130$ call is far out of the money and generally has a small premium.

1.3 Useful payoff combinations

Payoff algebra is often the quickest way to identify a trading strategy.

$$\begin{aligned} \text{long straddle:} &\quad (S_T - K)^+ + (K - S_T)^+ = |S_T - K|, \\ \text{protective put:} &\quad S_T + (K - S_T)^+ = \max(S_T, K), \\ \text{covered call:} &\quad S_T - (S_T - K)^+ = \min(S_T, K). \end{aligned}$$

Thus a straddle benefits from a large move in either direction; a protective put retains stock upside while imposing a floor K; a covered call caps stock upside at K.

2 Put–call parity

2.1 Replication identity

Let $D_t = e^{-r(T-t)}$ denote the price at time t of one dollar paid at time T, for a constant rate r. Consider a call and a put on the same stock, with the same K and T, and assume that the stock pays no dividends.

Proposition 2.1 (Put–call parity)**.** Their prices C_t and P_t satisfy

$$\boxed{C_t + KD_t = P_t + S_t,} \qquad \text{or equivalently} \qquad \boxed{C_t - P_t = S_t - Ke^{-r(T-t)}.}$$

Proof. Compare Portfolio A, one call plus cash KD_t, with Portfolio B, one put plus one share. At maturity their payoffs are

$$(S_T - K)^+ + K \qquad \text{and} \qquad (K - S_T)^+ + S_T.$$

If $S_T \le K$, both equal K; if $S_T > K$, both equal S_T. They are therefore the same terminal claim, namely $\max(S_T, K)$. The law of one price equates their values at time t. □

Parity is model-free

No distributional assumption, volatility parameter, or Black–Scholes formula appears in this proof. Parity follows from replication and a funding account. It is therefore one of the first consistency tests to apply to an option market.

Example — recovering a put price

Let $S_t = 100$, $K = 100$, $T - t = 1$, $r = 5\%$, and $C_t = 10.45$. Then

$$KD_t = 100e^{-0.05} \approx 95.12, \qquad P_t = C_t - S_t + KD_t \approx 10.45 - 100 + 95.12 = 5.57.$$

Knowing three prices in the parity relation determines the fourth.

2.2 Arbitrage from a parity violation

Suppose instead that

$$C_t + KD_t > P_t + S_t.$$

The call-plus-cash portfolio is expensive relative to the put-plus-stock portfolio. Sell the expensive portfolio and buy the cheap one. The initial receipt is strictly positive, while the net terminal payoff is exactly zero. The opposite inequality is handled by reversing the trade.

Exercise — diagnose a parity violation

Let $C_t = 12$, $P_t = 5$, $S_t = 100$, and $KD_t = 95.12$. Is parity satisfied? If not, give every leg of an arbitrage and its initial receipt.

Short solution

$$C_t + KD_t = 107.12 > 105 = P_t + S_t.$$

Sell the call, borrow 95.12, buy the put, and buy one share. The receipt is $107.12 - 105 = 2.12$ today. The two portfolios replicate the same payoff, so the terminal net payoff is zero.

Dividend adjustment

With a continuous dividend yield q, the stock leg has prepaid value $S_t e^{-q(T-t)}$, and parity becomes

$$C_t - P_t = S_t e^{-q(T-t)} - Ke^{-r(T-t)}.$$

Using the no-dividend formula for a dividend-paying stock can manufacture a spurious "arbitrage."

3 Black–Scholes as a pricing benchmark

3.1 The model and the formulas

Black–Scholes setting

Under the risk-neutral measure $\mathbb{Q}$, assume no dividends, constant rate r, constant volatility $\sigma > 0$, and

$$\frac{\mathrm{d}S_t}{S_t} = r\,\mathrm{d}t + \sigma\,\mathrm{d}W_t^{\mathbb{Q}}.$$

For $\tau = T - t$, the corresponding European call and put prices are

$$\boxed{C_t = S_t\Phi(d_1) - Ke^{-r\tau}\Phi(d_2)},$$

$$\boxed{P_t = Ke^{-r\tau}\Phi(-d_2) - S_t\Phi(-d_1)},$$

where

$$d_1 = \frac{\log(S_t/K) + (r + \frac{1}{2}\sigma^2)\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau}.$$

Here Φ is the standard normal distribution function. In this model, $\Phi(d_2) = \mathbb{Q}(S_T > K \mid \mathcal{F}_t)$, the risk-neutral probability that the call finishes in the money. The two formulas automatically satisfy put–call parity, as every arbitrage-free pricing model must.

Example — an at-the-money one-year call

For $S_t = K = 100$, $r = 5\%$, $\sigma = 20\%$, and $\tau = 1$,

$$d_1 = 0.35, \qquad d_2 = 0.15,$$

and, using $\Phi(0.35) \approx 0.6368$, $\Phi(0.15) \approx 0.5596$,

$$C_t = 100(0.6368) - 100e^{-0.05}(0.5596) \approx 10.45.$$

Parity then gives $P_t \approx 5.57$.

3.2 Greeks and local hedging

For an option price $V(S,t)$, the standard sensitivities are

Greek	notation	meaning
Delta	$\partial_S V$	sensitivity to a small stock move
Gamma	$\partial_{SS} V$	how quickly Delta changes
Vega	$\partial_\sigma V$	sensitivity to volatility
Theta	$\partial_t V$	sensitivity to passage of time
Rho	$\partial_r V$	sensitivity to the interest rate.

For a no-dividend Black–Scholes call,

$$\Delta_{\text{call}} = \Phi(d_1), \qquad \Gamma = \frac{\Phi'(d_1)}{S_t \sigma \sqrt{\tau}}, \qquad \text{Vega} = S_t \sqrt{\tau}\, \Phi'(d_1).$$

Delta is the number of shares used in a first-order hedge. Gamma means that this number changes as the stock moves; a Delta hedge therefore must be rebalanced.

Exercise — Delta hedge

A call has Delta 0.64. What is the first-order price change after a \$1 stock rise? How many shares hedge a short position in 100 calls?

Short solution

The call price rises by approximately \$0.64. The short-call position has Delta -64, so buying approximately 64 shares gives a locally Delta-neutral position. It is not risk-free: Gamma, volatility changes, and discrete rehedging remain.

3.3 Implied volatility: a surface in strike and maturity

Given a European call quote $C_t^{\text{mkt}}(K,T)$, its implied volatility is the value $\sigma_{\text{imp},t}(K,T)$ solving

$$C^{\text{BS}}(S_t, K, \tau, r, \sigma_{\text{imp},t}(K,T)) = C_t^{\text{mkt}}(K,T), \qquad \tau = T - t.$$

We hold spot, strike, maturity, and the interest rate fixed and invert the Black–Scholes formula with respect to σ. For $S_t, K, \tau > 0$, Black–Scholes Vega is strictly positive, so the solution is unique when it exists. A quote strictly between $(S_t - Ke^{-r\tau})^+$ and S_t has a unique positive, finite implied volatility, found numerically, for example by bisection. The earlier one-year call price of approximately 10.45, with $S_t = K = 100$ and $r = 5\%$, gives approximately 20%. Traders quote implied volatility because it makes options with different strikes and maturities easier to compare than their dollar premiums alone.

Two variables, at a fixed observation date

For a given underlying and a fixed date t, implied volatility is a **function of two variables: strike K and maturity T**:

$$\boxed{(K,T) \longmapsto \sigma_{\text{imp},t}(K,T).}$$

This is the **implied volatility surface**. Equivalently, use strike and time to maturity $\tau = T - t$. Each option quote supplies one point on the surface; there is generally no single implied volatility for a stock. The subscript t records when quotes are observed: the surface itself evolves as market conditions change.

There are two natural ways to take a slice of the surface:

- **Fix T, vary K:** obtain a smile or skew. A smile has higher implied volatilities in both wings than near the money; a downside skew has higher implied volatilities at lower strikes.
- **Fix K, vary T:** obtain a term structure of implied volatility. It can rise, fall, or be nonmonotone. Every volatility is annualized, regardless of the option's maturity.

Market displays may use moneyness instead of strike. An at-the-money term structure defined by $K = F_{t,T}$, for example, follows the forward at each expiry rather than holding the numerical strike fixed.

Example — reading two directions of the surface

At $S_t = 100$, consider the following illustrative annualized quotes:

Time to maturity	$K = 90$	$K = 100$	$K = 110$
3 months	28%	22%	20%
1 year	25%	21%	20%
2 years	24%	22%	21%

The first row shows a downside skew at a fixed 3-month maturity. The middle column shows a nonmonotone term structure at the fixed strike 100. Saying "implied volatility is 22%" is incomplete without specifying both strike and maturity.

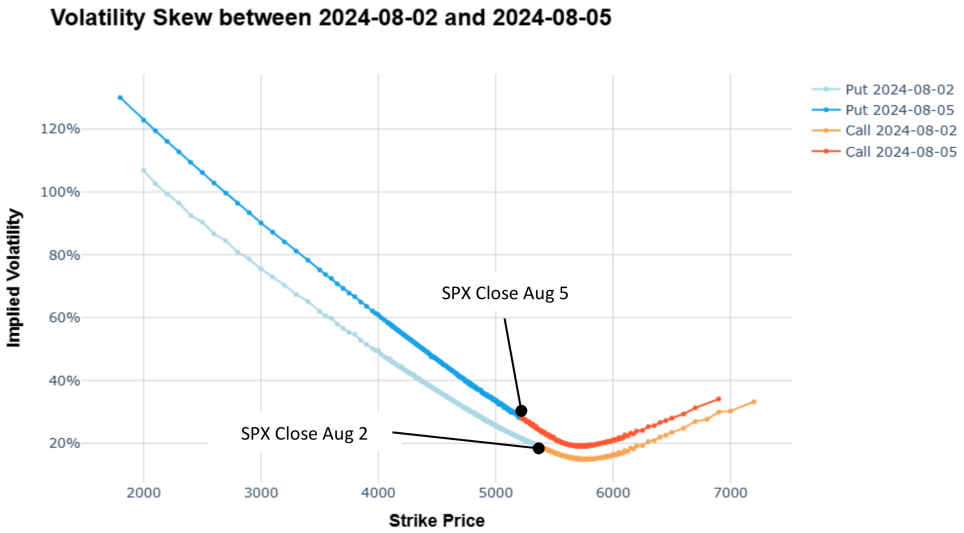
A single constant-volatility Black–Scholes model predicts the same implied volatility across *all strikes and maturities*: a flat surface. In equity markets, low-strike out-of-the-money puts often have higher implied volatilities than at-the-money options. Demand for crash protection can contribute to this downside skew. European calls and puts with the same (K, T) have the same implied volatility when their prices satisfy put–call parity and the same pricing inputs are used.

A price quotation, not a guaranteed forecast

Implied volatility is extracted from today's option price; historical realized volatility is estimated from past returns. Implied volatility is not a guaranteed forecast of future realized volatility. A nonflat surface shows that constant-volatility Black–Scholes is an imperfect description of market prices; a skew is not in itself an arbitrage.

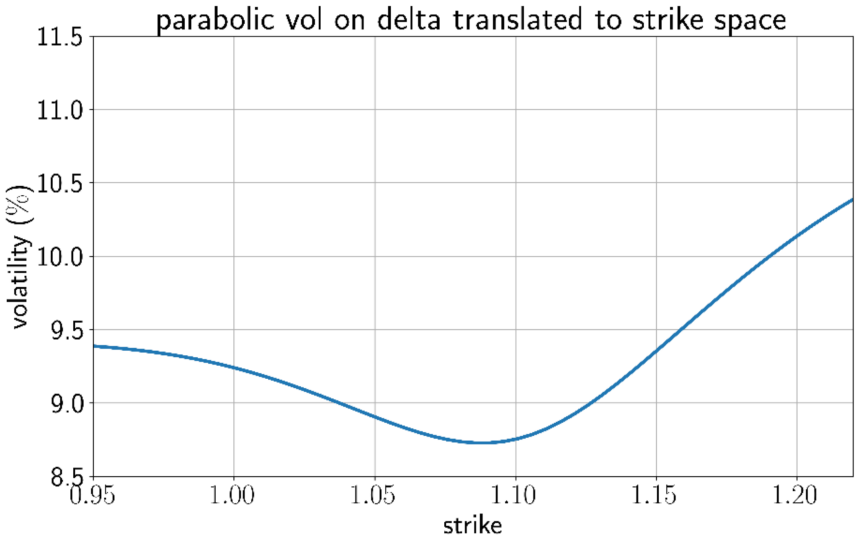
3.4 Market figures: downside skew and currency smile

The following published figures use historical market quotes. They show annualized implied volatility against strike; each curve has a fixed tenor.



S&P 500 options: downside skew. The two curves correspond to 2 and 5 August 2024, with a 30-day tenor interpolated in variance between expiries. Near the money, lower strikes carry higher implied volatilities. The right wing eventually rises too: skew and smile curvature can coexist.

Source: Cboe, Edward K. Tom, *The VIX Index Decomposition*, 1 August 2025, Exhibit 7, p. 9 (chart reproduced). The chart legend and surrounding text date the observations to 2024; the source's exhibit heading contains a year typo.



EUR/USD options: an asymmetric smile. For a six-month tenor on 23 April 2025, both wings lie above the trough. The strike is measured in USD per EUR. The smile is not symmetric, and its minimum need not coincide with the spot or forward.

Source: Uwe Wystup (MathFinance), *Why Digitals don't Have all Percentage Price Quotes*, 2025, Figure 2, p. 2; market quotes from ICE Data Services. The author uses parabolic interpolation in delta, converts it to strike, and fixes the reference spot at 1.1000 for illustration. This is a curve constructed from market quotes, not a scatterplot of individual trades.

4 Model-free bounds

4.1 Calls and puts lie in known intervals

Proposition 4.1 (Basic no-arbitrage bounds)**.** With no dividends and $D_t = e^{-r(T-t)}$,

$$\boxed{\max(S_t - KD_t, 0) \le C_t \le S_t,} \qquad \boxed{\max(KD_t - S_t, 0) \le P_t \le KD_t.}$$

Proof. For the call upper bound, one share superhedges the call because $S_T \ge (S_T - K)^+$. Thus a call cannot cost more than the share. If it did, sell the call, buy the share, and retain the positive difference today.

For the lower bound, call positivity gives $C_t \ge 0$, and parity gives $C_t = P_t + S_t - KD_t \ge S_t - KD_t$ because $P_t \ge 0$. Combining them gives the stated lower bound. The put bounds follow by the symmetric argument, or by applying parity to the call bounds. □

Exercise — a fast quote check

For $S_t = K = 100$, $r = 5\%$, and $\tau = 1$, classify call quotes 2, 10.45, and 103 as impossible or potentially arbitrage-free.

Short solution

The bounds are

$$\max(100 - 100e^{-0.05}, 0) \approx 4.88 \le C_t \le 100.$$

Thus 2 is too low, 103 is too high, and 10.45 is inside the model-free interval. Being inside the interval is necessary, not sufficient, for all cross-strike and cross-maturity consistency checks.

4.2 Superhedges and subhedges

The option bounds are instances of a general principle. If tradable portfolios have terminal values $Y_T \le H \le X_T$, their initial costs satisfy

$$\boxed{Y_0 \le \text{price}(H) \le X_0.}$$

If the claim is more expensive than the superhedge, sell the claim and buy the superhedge. If it is cheaper than the subhedge, buy the claim and short the subhedge. In both cases the trade pays today and cannot lose at maturity.

5 Strike and maturity consistency

5.1 Monotonicity in the strike

Proposition 5.1 (Strike monotonicity)**.** For the same underlying and maturity, if $K_1 < K_2$, then

$$C_t(K_1) \ge C_t(K_2), \qquad P_t(K_1) \le P_t(K_2).$$

Proof. For every terminal stock price,

$$(S_T - K_1)^+ \ge (S_T - K_2)^+.$$

The lower-strike call dominates the higher-strike call, so it cannot be cheaper. The put comparison reverses because the right to sell at a higher price is more valuable. □

Example — monotonicity arbitrage

Suppose $K_1 = 90 < K_2 = 110$, but $C_t(90) = 8 < C_t(110) = 10$. Buy the lower-strike call and sell the higher-strike call. The initial receipt is 2, while the terminal payoff

$$(S_T - 90)^+ - (S_T - 110)^+ \ge 0$$

in every state.

5.2 Convexity and butterfly spreads

Proposition 5.2 (Strike convexity)**.** For equally spaced strikes $K-h<K<K+h$, an arbitrage-free call curve satisfies

$$\boxed{C_t(K-h)-2C_t(K)+C_t(K+h)\geq 0.}$$

Equivalently, $K\mapsto C_t(K)$ is convex.

Proof. The butterfly payoff is

$$(S_T-K+h)^+-2(S_T-K)^++(S_T-K-h)^+.$$

It is zero outside $[K-h,K+h]$, increases linearly from zero to h, then decreases linearly to zero. It is therefore nonnegative in every state, so its price must be nonnegative. □

Exercise — a butterfly quote check

At equally spaced strikes $90,100,110$, calls are quoted at $14,12,7$. Check monotonicity and convexity. If necessary, construct the arbitrage.

Short solution

Monotonicity holds: $14>12>7$. Convexity requires

$$C_t(100)\leq \tfrac{1}{2}\big(C_t(90)+C_t(110)\big)=10.5,$$

but the quote is 12. Buy the 90 and 110 calls and sell two 100 calls. The initial cost is

$$14+7-2(12)=-3,$$

a receipt of 3, while the terminal butterfly payoff is nonnegative.

5.3 A qualified maturity comparison

If rates and dividends are zero and $T_1<T_2$, calls with the same strike satisfy

$$C_t(K,T_2)\geq C_t(K,T_1).$$

If this failed, buy the longer option and sell the shorter option. At T_1, the remaining longer option is worth at least the intrinsic value needed to settle the short call. This is the calendar-spread argument.

Do not overstate calendar monotonicity

Positive rates and dividends change the comparison across maturities. The simple inequality above is a useful clean case, not a universal replacement for the full calendar-arbitrage conditions used to construct option surfaces.

6 Synthesis and practice

A practical order for quote diagnostics

1. Match the contract specifications: underlying, K, T, and dividends.
2. Check positivity and the basic upper and lower bounds.
3. Check put–call parity for calls and puts with the same K,T.
4. Across strikes, check monotonicity and convexity.
5. Across maturities, use calendar checks with their stated assumptions.
6. When an inequality fails, write the portfolio and its cashflows before labelling the opportunity an arbitrage.

Exercise — final diagnostic

Assume no dividends. Given $S_t = 100$, $K = 100$, $r = 5\%$, $\tau = 1$, $C_t = 12$, and $P_t = 5$, identify the inconsistency and construct the trade.

Short solution

Since $100e^{-0.05} = 95.12$,

$$C_t + KD_t = 107.12 > 105 = P_t + S_t.$$

Sell the call, borrow 95.12, buy the put, and buy one share. The trade receives 2.12 initially and has zero terminal net payoff by put–call parity.

Final synthesis

$$\text{option payoff geometry} \quad \Longrightarrow \quad \text{arbitrage-free prices and hedges.}$$